

helix_code_services.md — Services Guide

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Overview

HelixCode service management wrappers: install, boot, stop, health-check, and status for the HelixCode / HelixLLM / HelixAgent service fleet.

Prerequisites

- `helix_code_home.sh` sourced first (provides `hc_home()`)
- bash $\geq$ 4.0
- Docker or Podman (for compose-based services)
- Network access to `github.com:HelixDevelopment/HelixCode.git`

Quick Start

```
source constitution/scripts/helix_code/helix_code_home.sh
source constitution/scripts/helix_code/helix_code_services.sh

# Install binaries
hc_install

# Boot the coder (multi-track live sink)
hc_boot_coder

# Check status
hc_status
```

```
# Verify all binaries on PATH
hc_verify_binaries
```

Functions

Function	Purpose
<code>hc_install [--flags]</code>	Install HelixCode binaries via <code>install_helix_path.sh</code>
<code>hc_boot_service <name></code>	Boot a service via the Containers submodule <code>compose</code>
<code>hc_boot_coder</code>	Boot the HelixLLM coder on <code>:18434</code> (with health wait)
<code>hc_stop_service <name></code>	Stop a service
<code>hc_status</code>	Print health status of every service in the <code>hc__status_endpoints</code> table
<code>hc_verify_binaries</code>	Verify all 6 binaries are on PATH
<code>hc_health [host] [port] [path]</code>	Health check over <code>http</code> only (from <code>helix_code_home.sh</code>)
<code>hc_health_url <url></code>	Scheme-aware health check; prints <code>OK</code> or <code>DOWN</code>

Service Ports

These are the rows of the `hc__status_endpoints` table in `helix_code_services.sh` — that table is the single source, this table mirrors it. Verified against the consuming project's `scripts/systemd/*.service` units and a live `ss -ltnp` on 2026-09-03.

Service	Scheme	Port	Health Endpoint
HelixCode server	http	:8080	/health
HelixLLM coder	http	:18434	/health
HelixLLM gateway	https	:8443	/internal/health
HelixAgent	http	:7061	/v1/models
LLMsVerifier	http	:8100	/health

Notes:

- **HelixAgent is :7061 , not :8100 .** :8100 is the **LLMsVerifier's** port; a prior revision of this table and of `hc_status` attributed it to HelixAgent, so `hc_status` reported "HelixAgent: DOWN" while HelixAgent was serving normally. HelixAgent also exposes a liveness probe on :8111 , but that endpoint answers 200 while the service is still starting, so the status probe deliberately targets `:7061/v1/models` — the port consumers actually call.
- **The gateway is https .** `hc_health` hardcodes `http://` , which is why `hc_health_url` exists; plain `http` against the TLS listener answers 400.

Safety

- §11.4.161: uses rootless podman via the Containers submodule
- §11.4.119: single-owner GPU — only one track boots the coder
- §12.6: 60% memory ceiling applies to service boot

Cross-references

- `docs/helix_code/INTEGRATION_PLAN.md` §3 Tasks 1.2-4
- `docs/helix_code/CLAUDE_TOOLKIT_EXTENSION.md` §1
- `docs/helix_code/RISK_ANALYSIS.md` — port collisions, GPU contention